

Functional form of $g(\omega)$

$$g(\omega) = \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma(h) e^{i\omega h}$$

$$= \frac{1}{2\pi} \left\{ \gamma(0) + \sum_{h=1}^{\infty} \gamma(h) e^{i\omega h} + \sum_{h=1}^{\infty} \gamma(-h) e^{-i\omega h} \right\}$$

$$= \frac{1}{2\pi} \left\{ \gamma(0) + \sum_{h=1}^{\infty} [\gamma(h) e^{i\omega h} + \gamma(-h) e^{-i\omega h}] \right\}$$

$$= \frac{1}{2\pi} \left\{ \gamma(0) + \sum_{h=1}^{\infty} \gamma(h) (e^{i\omega h} + e^{-i\omega h}) \right\}$$

since $\gamma(h) = \gamma(-h)$.

$$e^{i\theta} + e^{-i\theta} \equiv 2\cos\theta$$

$$\therefore g(\omega) = \frac{1}{2\pi} \left\{ \gamma(0) + 2 \sum_{h=1}^{\infty} \gamma(h) \cos(\omega h) \right\}$$

$$= \frac{\gamma(0)}{2\pi} + \frac{1}{\pi} \sum_{h=1}^{\infty} \gamma(h) \cos(\omega h)$$